

MODEL QUESTION PAPER

TED (10)-5074
(REVISION-2010)

Register Number.....
Signature.....

FIFTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY

POWER PLANT INSTRUMENTATION

(Elective for Instrument technology)

[Time: 3 Hours]
(Maximum marks: 100)

PART – A

I answer all questions.(Each question carries two marks.)

- 1) Write the principle of operation of Thermal power plant.
- 2) Differentiate water tube & fire tube boilers
- 3) State the importance of measurement of feed water temperature in boilers.
- 4) Write down the relevance of air/fuel ratio control in furnace
- 5) List any 4 air pollutants from thermal power plant. (5 x2 = 10 Marks)

PART – B

II Answer any five questions. Each question carries six marks.

- 1) Describe the non conventional methods of power generation
- 2) A coal conveyer system moves at a speed of 450 m/min. A weighing platform is 15 m in length and a particular weighing shows a reading of 200Kg. Find the coal delivery in Kg/hr.
- 3) Explain any one method for monitoring feedwater flow, with relevant schematic diagram
- 4) Describe the details of boiler-drum-level measurement with suitable diagram
- 5) Discuss how we can achieve the temperature control of reheat steam in boiler operation
- 6) Define thermal discharge index.
- 7) Draw the figure of a CO₂ analyser & explain its operation (5x6 = 30 Marks)

PART C

(Answer one full question from each module. Each question carries 15 marks.)

MODULE -I

- III(a) With the help of a schematic, explain the operation of a hydroelectric power plant (8)
(b) Describe the working of solar power generation (7)

OR

- IV (a) With the help of a neat diagram, explain the working of nuclear power plant (8)
(b) Write down the theory behind the operation of wind power plant (7)

MODULE –II

- V (a) Describe the methods for local and remote indications of drum water-level (15)

OR

- VI (a) Give the schematic of a typical demineralization plant and explain its operation (8)
b) Describe pressure measurement of boiler drum (7)

MODULE –III

- VII (a) Explain the schematic diagram of a steam turbine and explain its important functional parts (10)
(b) Write short note on pedestal vibration pick-up (5)

OR

- VIII (a) Distinguish between three element and five element boiler drum-level controls (10)
(b) Describe the purpose of combustion control in boiler operation (5)

MODULE –IV

- IX (a) Explain how an electrostatic precipitator can be used for particulate collection (10)
(b) List out the main air pollutants from thermal powerplant (5)

OR

- X (a) Draw the figure of an Orsat apparatus and explain its working (10)
(b) Discuss the operation of a thermal conductivity type hydrogen analyser (5)
